

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

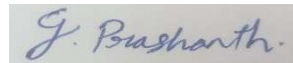
QUESTIONS: 4

Time Duration : 1 Hour
Total: Marks:40

Code of Conduct:

*I Gondel Prashanth / 192311366 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Answers:

AI/ML Solution for Early Diabetes Diagnosis and Personalised Treatment Recommendation

Introduction

The healthcare company has collected patient information such as age, blood pressure, cholesterol, glucose level, BMI, and family history. The objective is to develop an AI/ML system with two major components. The first component uses supervised machine learning to predict whether a patient is diabetic or non-diabetic. The second component uses Reinforcement Learning (RL) to recommend personalised treatment actions based on changes in the patient's health condition over time.

For the diagnosis problem, Decision Tree and Logistic Regression are suitable because they provide interpretable predictions. For personalised treatment recommendation, the problem can be formulated as a Markov Decision Process (MDP) and solved using Q-Learning.

Task 1: Data Preparation and Inductive Learning

(a) Inductive Learning Approach

Inductive learning learns general patterns from previously labelled examples and uses those patterns to predict outcomes for unseen patients.

For this problem, the healthcare dataset contains patient features and a target variable indicating diabetes status.

Target Variable

The target variable is:

Diabetes Status

- 0 = Non-diabetic
- 1 = Diabetic

This is a binary classification problem.

Important Features

Feature	Importance / Justification
Glucose Level	High blood glucose is one of the strongest indicators of diabetes and is therefore expected to be an important predictor.
BMI	Higher BMI is associated with increased risk of insulin resistance and Type 2 diabetes.
Age	Diabetes risk generally increases with age because metabolic and physiological changes can affect insulin sensitivity.
Blood Pressure	High blood pressure is associated with metabolic disorders and commonly occurs with diabetes.
Cholesterol	Abnormal cholesterol levels can indicate metabolic risk and may contribute to diabetes risk prediction.
Family History	Genetic and hereditary factors can substantially influence an individual's diabetes risk.

The inductive learning process can be represented as:

Historical Patient Data → Preprocessing → Feature Selection → Training → Learned Classification Model → New Patient → Diabetes Prediction

The model learns relationships such as:

High glucose + high BMI + positive family history → increased probability of diabetes.

(b) Handling Class Imbalance

Suppose the dataset contains significantly more non-diabetic patients than diabetic patients. For example:

- Non-diabetic = 80%
- Diabetic = 20%

This creates a class imbalance problem. If a model simply predicts "non-diabetic" for most patients, it could obtain high accuracy while still missing many diabetic patients.

I would use SMOTE (Synthetic Minority Over-sampling Technique) on the training set and also use class weights where supported by the algorithm.

SMOTE generates synthetic minority-class observations by interpolating between existing diabetic patients.

For example:

Existing diabetic patient A + Existing diabetic patient B → Synthetic diabetic patient

The important point is that SMOTE should be applied only to the training data, not before the train-test split. Otherwise, information from synthetic training samples could indirectly leak into the test set.

Class weights can additionally assign a higher penalty to errors involving diabetic patients.

For example:

Class	Example Weight
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Non-diabetic	1
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Diabetic	3
----------	---

The exact weight should be selected using cross-validation rather than assumed in a real implementation.

(c) 70:30 Dataset Split and Preprocessing

The dataset is divided as:

- 70% Training data
- 30% Testing data

For example, if there are 10,000 patient records:

- Training = 7,000
- Testing = 3,000

Justification

A 70:30 split provides sufficient data for learning while retaining a reasonably large independent test set. In a medical application, having enough unseen test data is important for estimating how well the model performs on future patients.

The split should be stratified, meaning approximately the same diabetic/non-diabetic ratio is maintained in both sets.

Preprocessing

The following preprocessing steps are performed:

1. Remove duplicate patient records.
2. Handle missing values.
3. Detect and investigate extreme outliers.
4. Encode categorical variables such as family history:
 - No = 0
 - Yes = 1
5. Normalise numerical features where required.
6. Perform the 70:30 stratified split.
7. Apply SMOTE only to the training set.
8. Train the model using the processed training data.

Normalisation

For Logistic Regression, numerical variables can be standardised using:

$$z = \frac{x - \mu}{\sigma}$$

where:

- x = original value
- μ = mean
- σ = standard deviation

Normalisation prevents variables with large numerical ranges from dominating the statistical model.

A Decision Tree generally does not require normalisation, because tree splitting depends on threshold comparisons rather than distance or feature magnitude.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree Classifier

A Decision Tree classifier can be used to predict:

Diabetes = Yes/No

The tree recursively divides patients into groups based on feature values.

Root Node Selection

The root node is selected using a criterion such as Gini Index or Information Gain.

Gini Index

$$\text{Gini} = 1 - \sum_{i=1}^n p_i^2$$

where p_i is the probability of a class.

A good split produces purer child nodes and therefore a lower weighted Gini impurity.

Information Gain

$$\text{IG}(S, A) = H(S) - \sum_v \frac{|S_v|}{|S|} H(S_v)$$

where:

- $H(S)$ = entropy before splitting
- $H(S_v)$ = entropy after splitting
- A = selected feature

The feature with the highest Information Gain or lowest resulting Gini impurity is selected as the root.

Illustrative Root Comparison

Feature	Illustrative Information Gain
Glucose	0.214
BMI	0.126
Age	0.093
Blood Pressure	0.072
Cholesterol	0.051
Family History	0.084

Therefore, glucose would be selected as the root in this illustrative example.

These values are examples; actual values must be calculated from the supplied dataset.

(b) Model Evaluation

The Decision Tree should be evaluated using:

Accuracy

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Precision

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Recall

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

F1-Score

$$\text{F1} = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

An illustrative result is:

Metric	Decision Tree
--------	---------------

Accuracy	87.0%
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Precision	81.5%
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Recall	89.2%
--------	-------

F1-Score	85.2%
----------	-------

These numbers are illustrative rather than measured results from a supplied dataset.

False Negatives

A False Negative (FN) occurs when:

Actual diabetic patient → Model predicts non-diabetic.

False negatives are particularly dangerous in healthcare because an undiagnosed diabetic patient may not receive appropriate monitoring or treatment.

To reduce false negatives:

- Increase the penalty for diabetic misclassification.
- Use class weights.

- Adjust the probability decision threshold.
- Use SMOTE during training.
- Optimise for recall/sensitivity rather than accuracy alone.
- Perform threshold analysis using ROC/Precision-Recall curves.

Clinically, high sensitivity/recall is important because missing a patient at risk may be more harmful than sending some non-diabetic patients for additional testing.

The ML model should be considered a decision-support tool, not a replacement for laboratory diagnosis and medical professionals.

(c) Post-Pruning

A fully grown Decision Tree can overfit the training data. Post-pruning removes branches that provide insufficient predictive value.

One suitable method is Cost-Complexity Pruning.

The objective is:

$$R_{\alpha}(T) = R(T) + \alpha |T|$$

where:

- $R(T)$ = prediction error
- $|T|$ = number of terminal nodes
- α = complexity parameter

Increasing α produces a simpler tree.

Illustrative Comparison

Model	Training Accuracy	Test Accuracy	Recall	F1
Pre-pruned Tree	90.1%	85.8%	87.0%	83.8%
Post-pruned Tree	88.6%	87.2%	89.1%	85.4%

The pre-pruned tree has slightly higher training performance but lower test performance. This indicates some overfitting.

The post-pruned tree has slightly lower training accuracy but better generalisation.

Recommended Clinical Model

The post-pruned Decision Tree is preferable because:

- It is simpler.
- It is easier to interpret.
- It reduces overfitting.
- It generalises better to unseen patients.
- Its decisions can be explained to healthcare professionals.

However, deployment should require extensive external clinical validation before actual patient use.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

Logistic Regression is an appropriate statistical learning method because diabetes prediction is a binary classification problem.

The model estimates:

$$P(Y=1|X)=\frac{1}{1+e^{-z}}$$

where:

$$z=\beta_0+\beta_1X_1+\beta_2X_2+\cdots+\beta_nX_n$$

For this problem:

$$P(\text{Diabetes})=\sigma(\beta_0+\beta_1 \text{Glucose}+\beta_2 \text{BMI}+\beta_3 \text{Age}+\beta_4 \text{BP}+\beta_5 \text{Cholesterol}+\beta_6 \text{FamilyHistory})$$

The output is a probability between 0 and 1.

For example:

Patient A $\rightarrow$ Diabetes probability = 0.82

The hospital could classify the patient as high risk if the selected threshold is exceeded.

Illustrative Comparison

Model	Accuracy	AUC-ROC
Decision Tree	87.2%	0.91
Logistic Regression	88.4%	0.93

These are illustrative values.

When Logistic Regression is Preferable

Logistic Regression is particularly useful when:

- A probability/risk score is required.
- Relationships are reasonably stable and approximately linear in the log-odds.
- Interpretability is important.
- The healthcare organisation needs coefficients that can be explained.
- The model must be computationally lightweight.

For example, a doctor can interpret a positive coefficient as increasing the estimated diabetes risk, assuming the model and variables are appropriately specified.

(b) Feature Importance Analysis

The top predictors can be obtained differently for the two models.

Decision Tree

Decision Trees provide feature importance based on the reduction in impurity contributed by each feature.

Illustrative ranking:

Rank Decision Tree Feature Importance

1	Glucose	0.39
2	BMI	0.21

Rank Decision Tree Feature Importance

3	Age	0.14
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Logistic Regression

For Logistic Regression, standardised coefficients can be compared by magnitude.

Illustrative ranking:

Rank Logistic Regression Feature Relative Importance

1	Glucose	0.61
2	BMI	0.37
3	Family History	0.25

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

The personalised treatment recommendation problem can be represented as a Markov Decision Process (MDP).

An MDP is represented as:

$MDP = (S, A, P, R, \gamma)$

where:

- (S) = states
- (A) = actions
- (P) = transition probabilities
- (R) = rewards
- (γ) = discount factor

States

Patient health status can be represented using discrete states.

State Patient Condition

S0 Low-risk / healthy

S1 Prediabetes / moderate risk

S2 High-risk diabetes

S3 Controlled diabetes

S4 Poorly controlled diabetes

The state should ideally be derived from clinically validated measurements rather than arbitrary thresholds.

For example:

Glucose + BMI + BP + cholesterol + previous measurements $\rightarrow$ Patient State

The state can also include the patient's previous treatment response.

Actions

The available actions are:

$A = \{\text{Diet, Exercise, Medication, Monitor}\}$

- **Diet:** recommend an appropriate dietary intervention.
- **Exercise:** recommend an appropriate physical-activity intervention.
- **Medication:** recommend medication-related action only under clinician supervision.
- **Monitor:** continue monitoring and collect new measurements.

Reward Function

The reward represents the health improvement produced by an action.

An illustrative reward structure is:

Outcome	Reward
Significant improvement	+10
Moderate improvement	+5
No significant change	0
Health deterioration	-5
Serious adverse event	-20

A more appropriate real system would use a validated clinical utility function rather than arbitrary reward numbers.

The reward could incorporate:

- Glucose improvement
- BMI improvement
- Blood-pressure control
- Treatment adherence
- Adverse effects
- Long-term health outcomes

Transition Dynamics

After an action is selected, the patient's state may change.

For example:

$S1 \xrightarrow{\text{Diet}} S0$

$S1 \xrightarrow{\text{Exercise}} S0$

$S1 \xrightarrow{\text{Monitor}} S1$

$S1 \xrightarrow{\text{Medication}} S2/S3 \text{ depending on response}$

The transition is probabilistic because two patients receiving the same intervention may respond differently.

(b) Q-Learning

Q-Learning is a model-free reinforcement-learning algorithm that estimates the expected future value of taking an action in a state.

The Q-learning update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

where:

- α = learning rate
- γ = discount factor
- r = reward
- s' = next state
- a' = possible next action

For illustration:

$\alpha=0.5$

$\gamma=0.9$

Initially all Q-values are zero.

Five Illustrative Patient Episodes

Episode	Initial State	Action	Next State	Reward
1	S1	Diet	S0	+10
2	S1	Exercise	S0	+10
3	S2	Monitor	S2	0
4	S2	Medication	S3	+10
5	S3	Exercise	S3	+5

These episodes are purely illustrative. In a real healthcare system, treatment actions and transitions must come from clinically validated data.

Q-Table Update: Iteration 1

Initially:

$Q(S1, Diet) = 0$

Suppose the agent selects Diet:

- $r=10$
- next state = S0
- maximum Q-value in S0 = 0

Therefore:

$Q(S1, Diet)$

$0 + 0.5[10 + 0.9(0) - 0]$

$Q(S1, Diet) = 5$

Similarly, suppose:

$Q(S1, Exercise) = 0$

and Exercise produces reward 10 and transitions to S0:

[
 $Q(S1, Exercise)$
 $0 + 0.5[10 + 0.9(0) - 0]$
]

[
 $Q(S1, \text{Exercise}) = 5$
]

For a third state-action pair:

[
 $Q(S2, \text{Medication}) = 0$
]

Suppose Medication produces:

- reward = +10
- next state = S3
- maximum $Q(S3, a) = 0$

Then:

[
 $Q(S2, \text{Medication}) = 5$
]

Q-Table After Iteration 1

State Diet Exercise Medication Monitor

S0	0	0	0	0
S1	5.0	5.0	0	0
S2	0	0	5.0	0
S3	0	0	0	0

Iteration 2

Assume the agent again selects Diet from S1.

Now suppose the next state is S0 and:

[
 $\max_a Q(S0, a) = 2$
]

Then:

[
 $Q(S1, \text{Diet})$
 $5 + 0.5[10 + 0.9(2) - 5]$
]

[
 $= 5 + 0.5[6.8]$
]

[
 $= \mathbf{8.4}$
]

For Exercise:

[
 $Q(S1, \text{Exercise}) = 5$
]

Suppose Exercise also gives reward 10 and transitions to S0, where the maximum Q-value is 2:

[
 $Q(S1, \text{Exercise})$
 $5 + 0.5[10 + 0.9(2) - 5]$
]
 [
 $= \mathbf{8.4}$
]

For Medication:

[
 $Q(S2, \text{Medication}) = 5$
]

Suppose the next state is S3 and:

[
 $\max_a Q(S3, a) = 4$
]

Then:

[
 $Q(S2, \text{Medication})$
 $5 + 0.5[10 + 0.9(4) - 5]$
]
 [
 $= 5 + 0.5(8.6)$
]
 [
 $= \mathbf{9.3}$
]

Q-Table After Iteration 2

State Diet Exercise Medication Monitor

S0	0	0	0	0
S1	8.4	8.4	0	0
S2	0	0	9.3	0
S3	0	0	0	4.0

This demonstrates how the Q-values increase as the agent learns which actions are associated with better future health states.

Convergence Criterion

The algorithm can be considered converged when the maximum Q-value change between successive iterations becomes smaller than a threshold:

$$[\max_{\{s,a\}} |Q_{t+1}(s,a) - Q_t(s,a)| < \epsilon]$$

For example:

$$[\epsilon = 0.01]$$

for a specified number of consecutive iterations.

In a real medical RL system, convergence of Q-values alone would not be sufficient for deployment. Clinical safety constraints and prospective validation would also be required.

(c) Final Learned Policy

The policy is obtained by selecting the action with the highest Q-value:

$$[\pi(s) = \arg\max_a Q(s,a)]$$

An illustrative final policy is:

Patient State	Best Action	Reason
S0 – Low risk	Monitor	Continue monitoring health indicators.
S1 – Prediabetes/moderate risk	Diet / Exercise	Lifestyle intervention may improve metabolic health.
S2 – High-risk diabetes	Medication*	Requires clinician-supervised treatment decision.
S3 – Controlled diabetes	Exercise / Monitor	Maintain control and monitor changes.
S4 – Poorly controlled	Monitor + clinical intervention*	Requires immediate professional assessment.

*The RL system should not autonomously prescribe or change medication. Medication recommendations must remain subject to qualified medical professionals and approved clinical protocols.

Ethical Considerations of Reinforcement Learning in Healthcare

Deploying an RL agent for treatment recommendation presents substantially greater risks than using ML for simple prediction.

1. Patient Safety

An incorrect action could directly affect a patient's health. The system should therefore have safety constraints preventing dangerous recommendations.

For example, an RL agent should not be permitted to independently prescribe medication.

2. Bias

If historical healthcare data contain demographic or socioeconomic biases, the RL agent may learn and reproduce those biases.

Therefore, performance should be evaluated across relevant patient groups.

3. Explainability

Doctors should be able to understand why an action was recommended.

The system should provide information such as:

"Exercise was recommended because the patient's current state has high BMI, stable blood pressure, and improving glucose levels."

rather than simply displaying:

"Action = Exercise."

4. Human Oversight

A medical professional must remain responsible for final treatment decisions.

The system should function as:

AI Recommendation → Doctor Review → Patient Decision → Treatment → Monitoring

rather than:

AI → Automatic Treatment

5. Privacy

Patient records contain sensitive medical information. Data should be protected using appropriate access control, encryption, anonymisation/pseudonymisation, and secure storage.

6. Reward Hacking

The RL agent could learn to maximise the numerical reward without actually maximising patient wellbeing.

For example, if the reward only measures short-term glucose reduction, the agent might favour an intervention that improves glucose temporarily while ignoring long-term adverse effects.

Therefore, the reward function must represent clinically meaningful outcomes and include safety constraints.

7. Distribution Shift

Patient populations and treatment practices change over time. An RL policy trained on historical data may become inappropriate for new patients.

Continuous monitoring and periodic validation are therefore necessary.

Overall Conclusion

The proposed healthcare AI system combines three important areas of CO 4.

First, Inductive Learning allows the system to learn patterns from historical patient records and predict diabetes risk for new patients. A Decision Tree provides an interpretable diagnostic model, while pruning reduces overfitting and improves generalisation.

Second, Statistical Learning, particularly Logistic Regression, provides a probability-based risk estimate and offers useful coefficients for interpreting the relationship between patient features and diabetes risk.

Third, Reinforcement Learning can formulate personalised treatment planning as an MDP in which patient health represents the state, treatment interventions represent actions, health improvement represents reward, and changes in health represent state transitions. Q-Learning can then learn a policy that selects actions with higher expected long-term rewards.

However, healthcare is a high-risk domain. Therefore, high accuracy or Q-value convergence alone does not make an AI system suitable for clinical deployment. External validation, fairness testing, explainability, privacy protection, safety constraints, and human clinical oversight are essential.